

Project 2 - Part 1: Linearization of the Shallow-Water Equations

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Q1) Linearization around the rest state

Starting equations

The shallow-water equations (SWE) without drag are:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -g'\nabla h + \nu \nabla^2 \mathbf{u}. \quad (2)$$

Step 1: Write each variable as mean + small perturbation

We assume the fluid is almost at rest, so we split each variable into a constant mean value and a small disturbance:

$$h(x, y, t) = \bar{h} + \delta h(x, y, t), \quad (3)$$

$$\mathbf{u}(x, y, t) = \mathbf{0} + \delta \mathbf{u}(x, y, t). \quad (4)$$

The mean state is the rest state: $\bar{h}$ is constant (no gradient) and the mean velocity is zero. The perturbations are small:

$$|\delta h| \ll \bar{h}, \quad |\delta \mathbf{u}| \ll \sqrt{g'\bar{h}}.$$

Step 2: Drop terms that are products of two small quantities

If two small things multiply each other, the result is much smaller than the small things themselves. So we keep only terms that are linear in the perturbations and throw away quadratic terms.

Mass equation. Plug the split into eq. (1):

$$\frac{\partial(\bar{h} + \delta h)}{\partial t} + \nabla \cdot ((\bar{h} + \delta h) \delta \mathbf{u}) = 0.$$

Since $\bar{h}$ does not depend on time, $\partial \bar{h} / \partial t = 0$. Expand the divergence:

$$\frac{\partial \delta h}{\partial t} + \bar{h} \nabla \cdot \delta \mathbf{u} + \nabla \cdot (\delta h \delta \mathbf{u}) = 0.$$

The last term is quadratic in small quantities ($\delta h \cdot \delta \mathbf{u}$) so we drop it:

$$\boxed{\frac{\partial \delta h}{\partial t} + \bar{h} \nabla \cdot \delta \mathbf{u} = 0.} \quad (\text{A})$$

Momentum equation. Plug into eq. (2):

$$\frac{\partial \delta \mathbf{u}}{\partial t} + (\delta \mathbf{u} \cdot \nabla) \delta \mathbf{u} = -g' \nabla (\bar{h} + \delta h) + \nu \nabla^2 \delta \mathbf{u}.$$

The advection term $(\delta \mathbf{u} \cdot \nabla) \delta \mathbf{u}$ is quadratic so we drop it. Also $\nabla \bar{h} = 0$ because $\bar{h}$ is constant. We also ignore the viscous term $\nu \nabla^2 \delta \mathbf{u}$ because gravity waves are inviscid at leading order (viscosity gives damping, not the wave itself). What remains:

$$\boxed{\frac{\partial \delta \mathbf{u}}{\partial t} = -g' \nabla \delta h.} \quad (\text{B})$$

Step 3: Combine the two equations into one for δh

Take the time derivative of equation (A):

$$\frac{\partial^2 \delta h}{\partial t^2} + \bar{h} \nabla \cdot \frac{\partial \delta \mathbf{u}}{\partial t} = 0.$$

Now substitute equation (B) inside:

$$\frac{\partial^2 \delta h}{\partial t^2} + \bar{h} \nabla \cdot (-g' \nabla \delta h) = 0.$$

Pull the constants out and use $\nabla \cdot \nabla = \nabla^2$:

$$\frac{\partial^2 \delta h}{\partial t^2} - g' \bar{h} \nabla^2 \delta h = 0.$$

This matches the target equation (with $g \rightarrow g'$ as the reduced gravity used in the project):

$$\boxed{\frac{\partial^2 \delta h}{\partial t^2} - g' \bar{h} \nabla^2 \delta h = 0.} \quad (5)$$

Q2) Interpretation: why this is a gravity wave equation

The result has the form of the classical **wave equation**:

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \nabla^2 \phi = 0,$$

where the wave speed is

$$c_0 = \sqrt{g' \bar{h}}.$$

This tells us several things:

- Small perturbations of the surface height δh do not just sit still or decay; they propagate as waves through the fluid layer.
- The restoring force is gravity (via g'): if the surface bulges up, gravity pulls it back down; the inertia of the fluid makes it overshoot, and the disturbance travels.

- The speed depends on the mean depth $\bar{h}$: deeper layer means faster waves. This is exactly what shallow-water gravity waves do (tsunamis are the famous example).
- For the project parameters $g' = 0.3 \text{ m/s}^2$ and $\bar{h} = H_0 = 1000 \text{ m}$, we get $c_0 = \sqrt{0.3 \times 1000} \approx 17.3 \text{ m/s}$.

So the SWE *support gravity waves* because the linearized system is a wave equation with speed $c_0 = \sqrt{g'\bar{h}}$ driven by gravity acting on height disturbances.